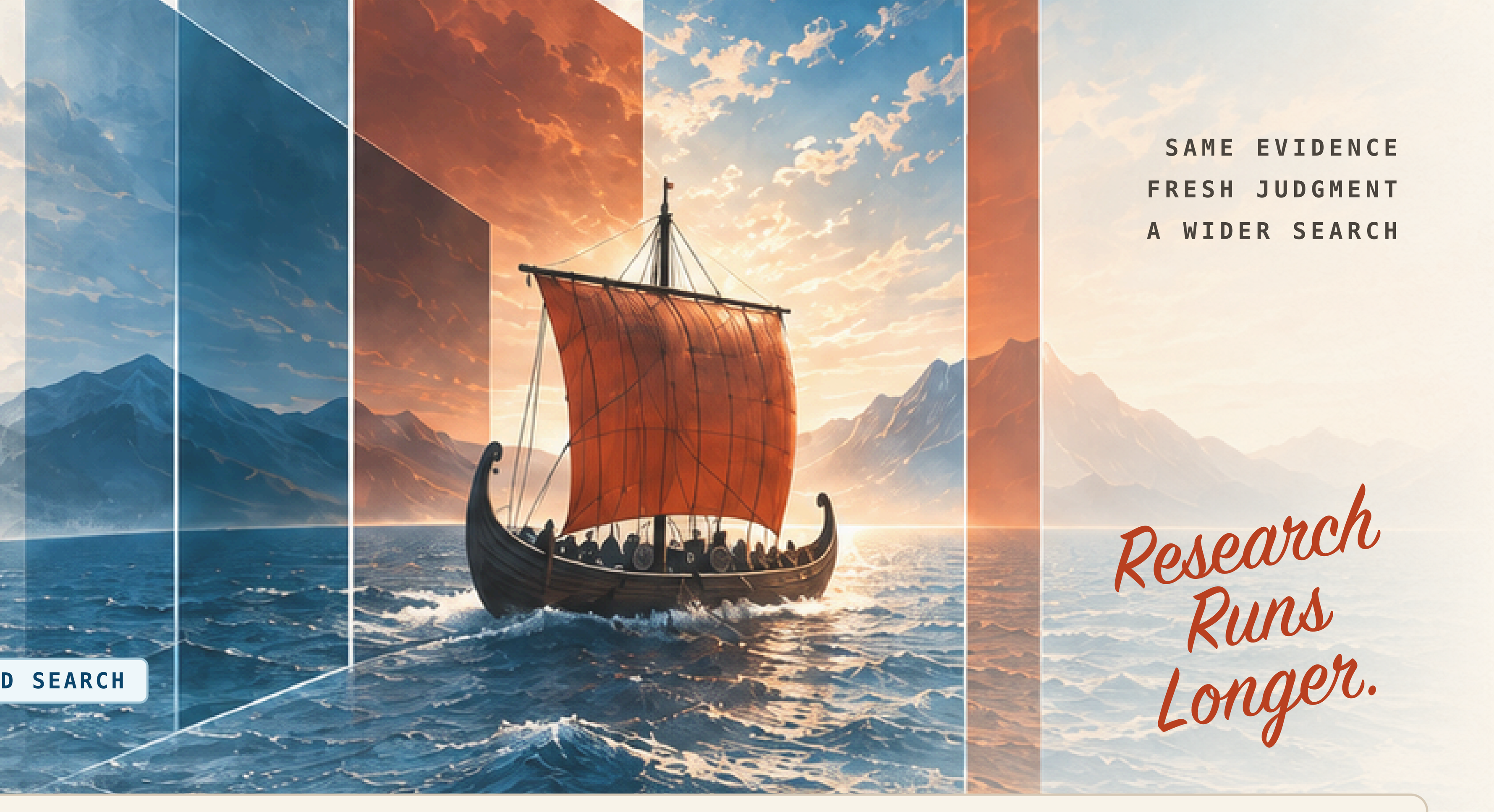
SciOdyssey.



A research layer over your agent harness.

— PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST · SUPERVISABLE · BROAD SEARCH

01 / PROBLEM

Why current agents fall short.

Sustained autonomous research breaks in three different ways.

01 Closed-world fragility

Unexpected runtime failures can halt an unattended workflow.

`Cycle 1 · numpy.float32 serialization error · autonomously repaired`

02 Trajectory momentum

A single long-running trajectory keeps moving in the same direction.

`Direct Codex control reached ~0.6046 after extended local refinement`

03 Context inflation cost

Reconstructing project context can cost far more than the edit itself.

`Project-shaped replication · 340,087 vs 42,377 inclusive tokens · 8.03×`

DESIGN RESPONSE Keep the world. Reset the trajectory.

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP — Persistent world · ephemeral trajectory · audited write-back

INPUTS: TASK · DATA · EVALUATOR · BUDGET

Fresh Scientist — Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

→ **META Review** — Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE — Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD — What survives across trajectories

- **A · EVIDENCE — Knowledge + Ledger**: Metrics, failures, reports, and experiment receipts in `ledger.jsonl`.
- **B · PRIORS — Brief + Hypotheses**: Current understanding and exploration leads in `brief.md`, open to revision.
- **C · STATE — Code + State**: Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET — The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

- **WORKTREE ISOLATION** — Independent git worktrees isolate branches and preserve reproducible artifacts.
- **SELF-HEALING** — Traceback → patch → rerun keeps unattended research moving after runtime faults.
- **PARALLEL SYNTHESIS** — Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not **prior reasoning momentum**.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT — The model got better.

0.6016000 (Official FM baseline) → **0.6059363** (SciOdyssey submission)

+0.0043363 Primary improvement

GAUC **0.6728421** (+0.0054) · nDCG@5 **0.5390304** (+0.0033) · LogLoss **0.38102** (-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms
Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

02 SUSTAINED SEARCH — The search stayed productive.

0.60163 (C1: Valid Base) — 0.60409 (C2: Features) — 0.60443 (C3: Var. Red.) — 0.60594 (C4: 8-Seed FM)

SciOdyssey	0.6059363	Climbed across 4 cycles (7 Full evals)
Codex Goal	0.6044533	Peaked at C8; stalled & dropped in 22 runs

4 autonomous cycles · **7** Full evaluations · **+0.00148** over Codex

7 Full public-validation evaluations retained across 4 cycles. Trajectory climbed via **structural feature expansion** rather than single lucky edits.

03 SELF-HEALING & GATEKEEPING — The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING — CRASH AUTO-RECOVERY
FAILURE: Cycle 1 `numpy.float32` serialization crash
ACTION: Scientist parsed traceback, auto-patched scalar helper
OUTCOME: **0 human bugfixes** · Unattended rerun succeeded

TIER 2 · META GATEKEEPING — SPURIOUS STATE REJECTION
FALSE CLAIM: Scientist claimed BPR gain on medium proxy
ACTION: META audited `system/data.py`, caught UNK leak
OUTCOME: **Promotion blocked** · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

04 ARCHITECTURE ABLATION — Meta-scientist beats direct agents.

01 · DIRECT — Antigravity — **0.60458** — Single 2h run
→ 02 · SUBAGENTS — + Subagents — **0.60472** — Delegated swarm
→ 03 · META-SCIENTIST — Persistent World — **0.60520** — + fresh Scientist

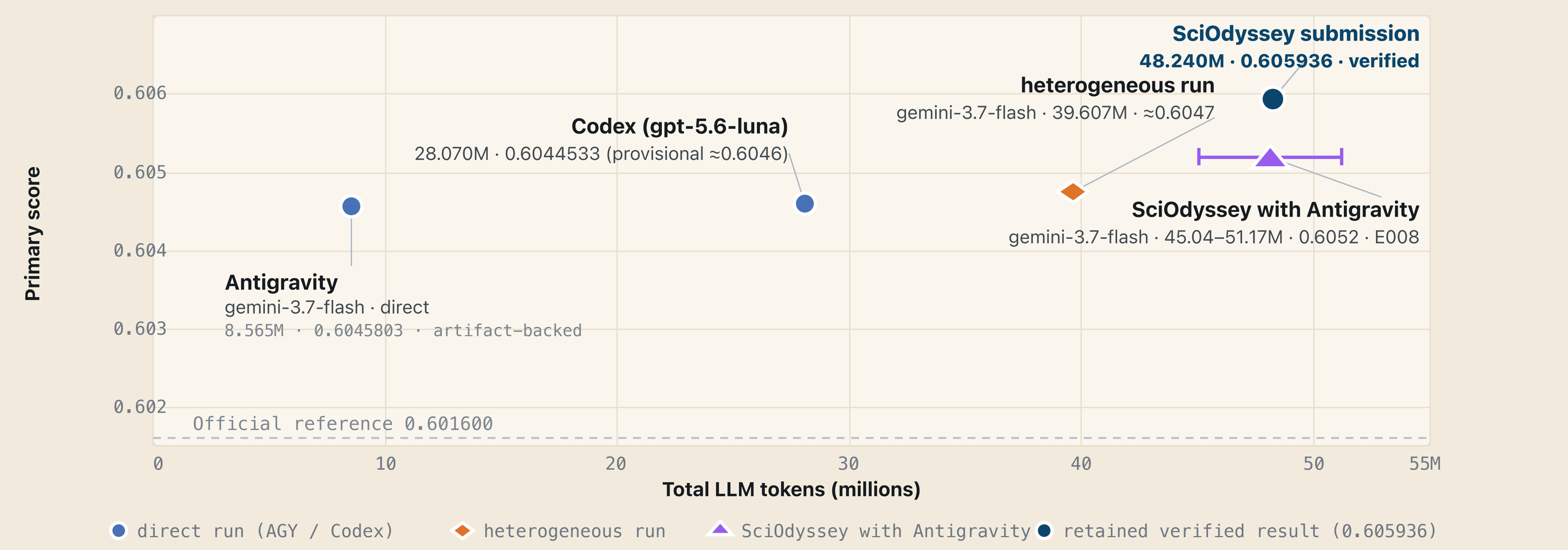
+0.00062 organization gain (Gemini 3.7 Flash)

Keep Gemini 3.7 Flash fixed and change only research organization: **direct agent (0.6046) → subagents (0.6047) → Meta-Scientist (0.6052)**.

05 RESOURCE ACCOUNTING — We are not tokenmaxxing. Score vs token investment.

~$10 cost · 1 × M2 · 0 GPU-hr · 1h 55m

KuaiRand-Pure · Public validation · Inclusive input+output tokens

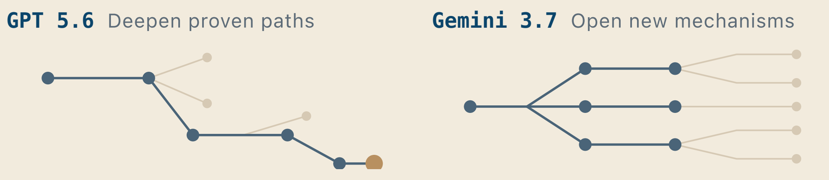


All results **verified on KuaiRand-Pure official validation split** with **complete reproducible artifacts**.

04 / FINDINGS

What the run revealed.

01 · MODEL DIVERSITY — Model diversity can help break plateaus.



Observed rotation: GPT → Gemini → GPT → Gemini
↻ shared research world

Gemini-only **0.6052** → GPT+Gemini **0.6059363**

02 · PARALLEL SYNTHESIS — A losing branch can still improve the winner.

- A **0.6048** Redundant
- B **0.6048** Chosen ✓
- C **0.6039** Useful ideas

→ Reviewer (Synthesize B+C) → Fresh Scientist **0.6055536**

Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.

Branch B+C Ideas → Synthesis **0.6055536**

05 / EXPERIENCE

Start your journey.

Autonomous by default. Supervisable by design.

task.md — What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md — How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS — **Tier 1** Unattended Overnight Runs · **Tier 2** Checkpoint Review (META) · **Tier 3** Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL — **Human defines boundary and preferences** (task.md + PERSONAL.md); **SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.**

Coding agent harness · Execution studio — Works with: Codex · Claude Code · Antigravity · OpenCode · Trae

```
› SKILL.md "Use the research-agent skill. Explore in parallel; let me review what to keep."

$ research-agent step                                  # one cycle · human reviews
  META → Scientist → Evidence → Your turn ✓

$ research-agent run --max-cycles 4                    # autonomous · fresh scientist each round
  C1 ↺ → C2 ↺ → C3 ↺ → C4 · Candidate Retained ✓

$ research-agent parallel --branches 3                 # isolated worktrees · reviewer synthesizes
  Branch A·B·C → Reviewer Synthesis → Explicit Promotion ✓

$ research-agent gui                                   # dashboard · telemetry & checkpoints
  Telemetry · Loss Curves · Verified Checkpoints ✓
```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